

Quality Assurance of Large Language Model Based Medical Application

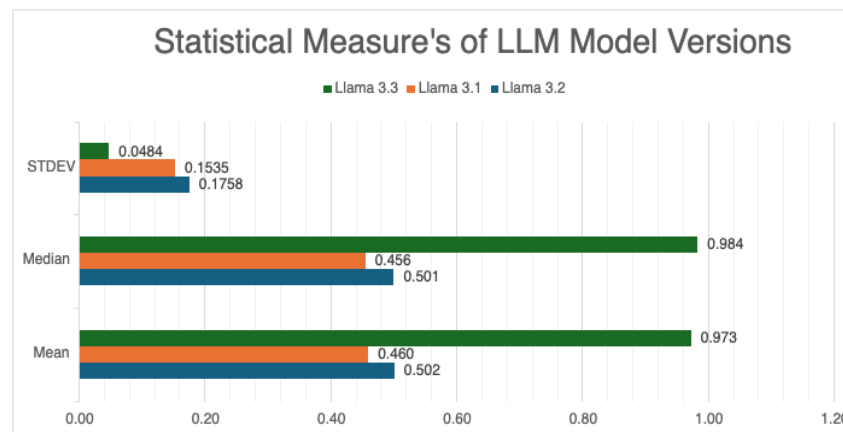
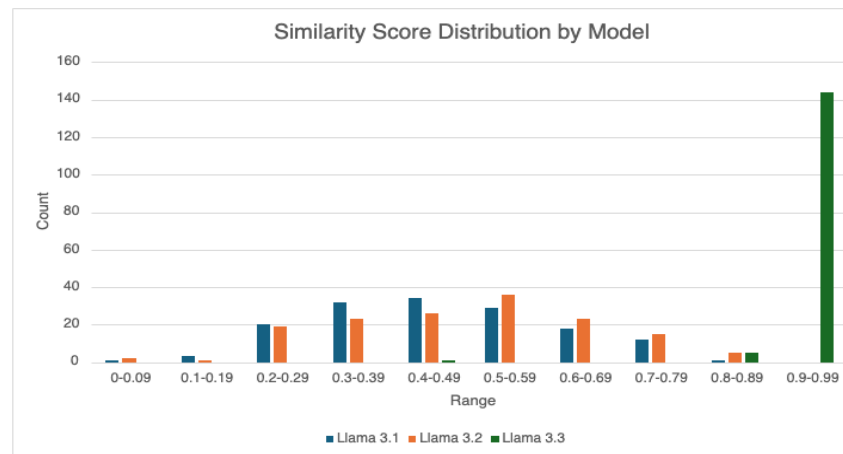
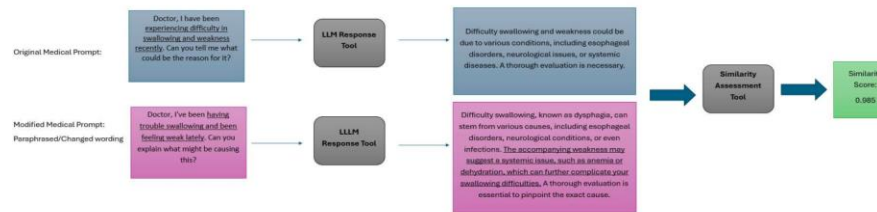
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INTRODUCTION:

The integration of technology in healthcare, particularly through the use of large language models (LLMs) and medical chatbots, has become increasingly prevalent. It was reported in 2024 that at least one fifth of healthcare workers used large language models (LLMs) and medical chatbots to answer patient questions and assess their symptoms. Often, Machine Learning models such as LLMs are developed with the focus of achieving the best possible accuracy. However, to use them in the medical domain, assessing quality attributes such as robustness and fairness is equally essential. This project deals with evaluating the robustness and fairness of LLMs when they are used in a medical context. The results of this project can shed light on how safe these LLMs are to be used in the medical domain by pointing out where they fail in terms of these quality attributes.

OBJECTIVES:

- Identify a suitable method for evaluating robustness in LLMs.
- Investigate methodologies for testing and utilizing large language models (LLMs)
- Select suitable LLM for testing and automate code using python
- Aggregate results, perform statistical analysis, and visualize findings through graphs
- Synthesize findings and report conclusions



PROCEDURE:

Large Language Models

The large language models (LLMs) utilized in this study were sourced from the Ollama framework, specifically employing three versions of the Llama model: Llama 3.1, Llama 3.2, and Llama 3.3. The example illustrated above demonstrates the output generated by Llama 3.3, the most recent and advanced version used in this study.

Data

To evaluate the performance and consistency of the models, a dataset comprising of 150 medically oriented prompts was compiled. Each original prompt was paired with a paraphrased version. This dataset was designed to assess the models' capabilities in understanding, responding to, and maintaining consistency across semantically similar inputs. The LLMs were tested on their ability to generate coherent and contextually appropriate responses to both the original and paraphrased prompts, providing insight into their robustness in handling linguistic variation.

CONCLUSIONS:

The testing showed that as model versions got more updated and advanced so did their responses. Overall the scores of the testing were above a 50% similarity with the higher values being from the most recent model Llama 3.3. The response's similarity scores from Llama 3.3 shows that using paraphrased versions can still predict the same result as the original prompt. This shows robustness of the models because they can handle the level of input and output necessary for a hundred and fifty prompts. Over 85% of results from Llama3.3 fell in the 0.90-0.99 range. This highlights the models updated ability to give back answers of high similarity between original and paraphrased prompts.